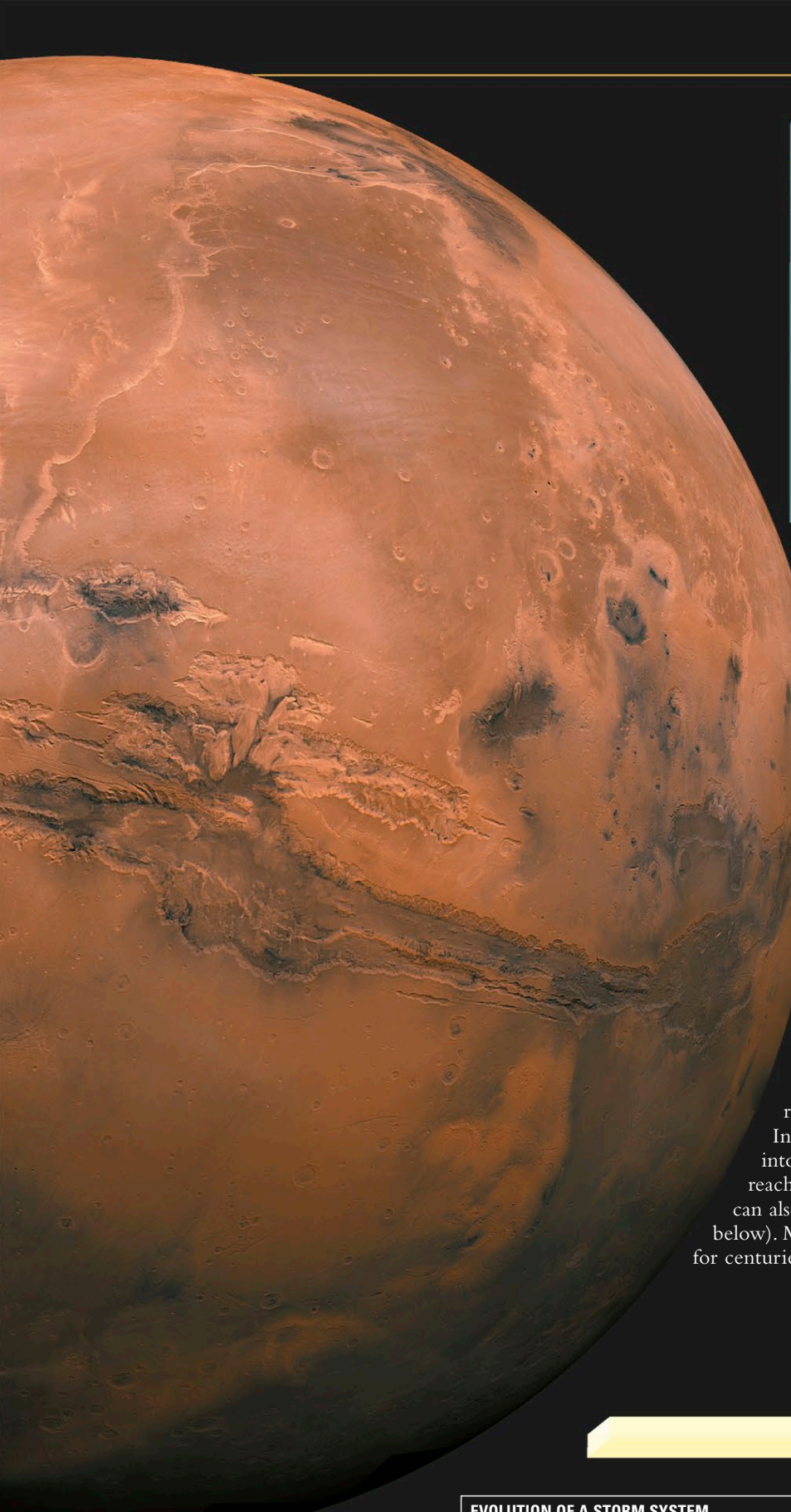




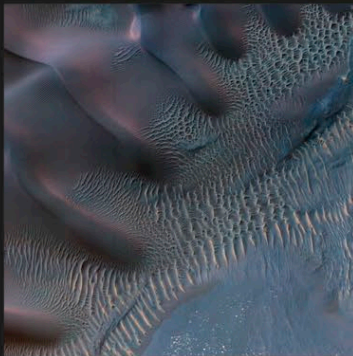
MARS PROFILE



AVERAGE DISTANCE FROM THE SUN 141.6 million miles (227.9 million km)	ROTATION PERIOD 24.63 hours
SURFACE TEMPERATURE -195°F to 77°F (-125°C to 25°C)	ORBITAL PERIOD (LENGTH OF YEAR) 687 Earth days
DIAMETER 4,213 miles (6,780 km)	MASS (EARTH = 1) 0.11
VOLUME (EARTH = 1) 0.15	GRAVITY AT EQUATOR (EARTH = 1) 0.38
NUMBER OF MOONS 2	SIZE COMPARISON
OBSERVATION Mars is visible to the naked eye. It is brightest when at its closest to Earth, which is approximately once every 2 years. It then has an average magnitude of -2.0.	EARTH  MARS 

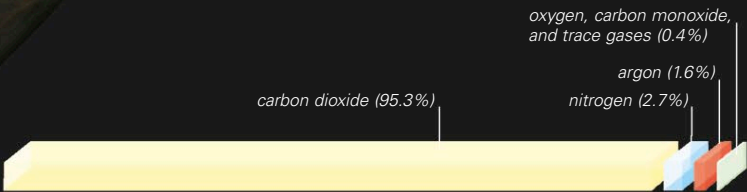
ATMOSPHERE AND WEATHER

Mars has a very thin atmosphere, which exerts an average pressure on the surface of about 6 millibars (0.6 percent of the atmospheric pressure on Earth). The atmosphere is mostly carbon dioxide, and it appears pink because fine particles of iron oxide dust are suspended in it. Thin clouds of frozen carbon dioxide and water ice are present at high altitudes, and clouds also form on high peaks in the summer. Mars is a cold, dry planet—the average surface temperature is -81°F



SAND DUNES
Looking down into a small impact crater in a southern upland area called Noachis Terra, NASA's Mars Reconnaissance Orbiter captured these rippling sand dunes. The dunes were sculpted by Martian winds and are shown here in enhanced color. The image is about 0.6 miles (1 km) across.

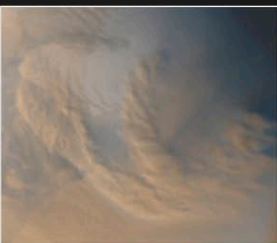
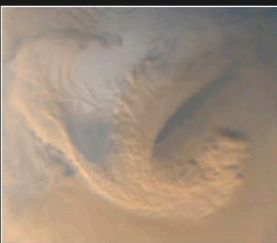
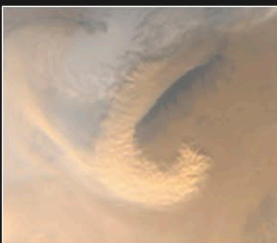
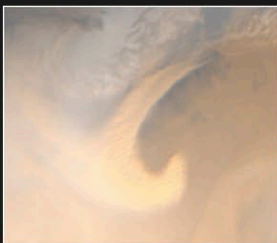
(-63°C)—where it never rains, but in the winter clouds at the polar regions cause ground frosts. Mars has highly dynamic weather systems. In the southern spring and summer, warmer winds from the south blow into the northern hemisphere, stirring up local clouds of dust that can reach 3,000 ft (1,000 m) in height and last for weeks. The high-level winds can also create powerful dust storms that cover vast areas of the planet (see below). Mars also has low-level prevailing winds that have sandblasted its surface for centuries, creating distinctive landforms (see photograph, above).



ATMOSPHERIC COMPOSITION
The thin atmosphere of Mars is dominated by carbon dioxide, with tiny amounts of nitrogen and argon and other gases and some traces of water vapor.

SCARRED SURFACE
This mosaic of Viking Orbiter images shows Mars's distinct red coloration and reveals the vast extent of the Valles Marineris, a system of valleys more than 2,500 miles (4,000 km) long.

EVOLUTION OF A STORM SYSTEM



1 On June 30, 1999, a storm system developed over the north polar region of Mars.

2 A giant, turbulent cloud of orange-brown dust was raised by high surface winds.

3 Expanding rapidly, the storm swirled over the white ice cap (center, top).

4 Six hours after the first image was taken, the storm was still gathering strength.